

New Case Study: SmartRetail Sdn. Bhd.

SmartRetail Sdn. Bhd. is a rapidly growing Malaysian retail company that operates physical stores and an online shopping platform. To improve customer experience and operational efficiency, the company has introduced several digital technologies, including **AI-powered recommendation systems, facial recognition cameras, mobile payment, cloud computing, Big Data analytics, and Internet of Things (IoT) devices.**

The company collects customer information such as purchase history, location data, browsing behaviour, and shopping preferences. AI is used to recommend products and provide personalised promotions.

Recently, SmartRetail introduced facial recognition technology at selected stores to identify returning customers and provide personalised discounts. However, some customers were unaware that their faces were being scanned.

The company also uses IoT sensors to monitor product inventory and track the movement of goods within warehouses. While this has improved inventory management, employees have raised concerns that the same technology is being used to monitor their productivity throughout the working day.

SmartRetail stores customer data on a cloud platform operated by a third-party provider. The company experienced a security incident where some customer accounts were accessed using stolen login credentials.

To reduce costs, management is also considering replacing customer service staff with AI-powered chatbots. Some employees are concerned that this could result in job losses and make it more difficult for customers with complex problems to receive human assistance.

The company has also introduced an algorithm that determines which customers receive promotional discounts. Some customers noticed that different people were receiving different prices for similar products and questioned whether the system was fair.

KBAT Questions

Q1. Artificial Intelligence & Employment

SmartRetail plans to replace some customer service employees with AI-powered chatbots.

Evaluate the ethical implications of replacing human employees with AI and recommend TWO measures SmartRetail could take to balance business efficiency with employee and customer interests.

Q2. Facial Recognition & Privacy

SmartRetail uses facial recognition to identify returning customers and provide personalised discounts, but some customers were unaware that their faces were being scanned.

Analyse the ethical concerns associated with this practice and evaluate whether SmartRetail should continue using facial recognition technology. Justify your answer.

Q3. Big Data & Algorithmic Bias

SmartRetail uses customer purchasing and browsing data to determine personalised promotions. Some customers receive different discounts for similar products.

Evaluate the fairness of using algorithms to determine customer discounts. What measures should SmartRetail implement to prevent discrimination or unfair treatment?

Q4. IoT & Employee Monitoring

SmartRetail uses IoT sensors to monitor employee movement and productivity throughout the working day.

Assess whether the use of IoT technology for employee monitoring is ethically justified. In your answer, consider both the company's interests and employees' privacy rights.

Q5. Cloud Computing & Data Security

Customer information is stored on a third-party cloud platform. Following a security incident involving stolen customer credentials, management is considering whether to continue using cloud computing.

Evaluate whether SmartRetail should continue using cloud services. Recommend THREE measures that could reduce the risks associated with storing customer information in the cloud.